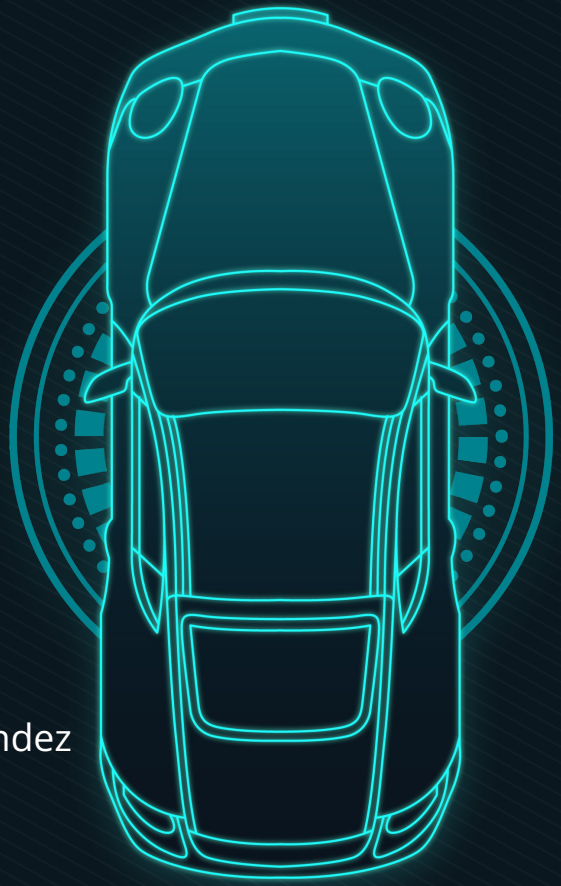
Survival Analysis of Vehicle Fuel Type and Longevity

Kayle Megginson, Cameron Hayman, Brandon Rodriguez-Hernandez



Questions of Interest

- How long do vehicles remain on the road, and how does this vary by fuel type?
- Do certain fuel types (e.g., diesel, midgrade, premium) lead to longer or shorter vehicle survival?
- What vehicle characteristics (e.g., engine size, fuel efficiency, emissions rating) are associated with longevity?
- Which survival model best fits the data (e.g., Weibull, exponential, Cox PH)?

Background

Understanding Vehicle Longevity in a Changing Automotive Landscape

A Landscape in Transition

- Traditional Factors
  Longevity has historically been influenced by:
  - Fuel type (e.g., diesel vs. gasoline)
  - Engine size and build quality
  - Owner maintenance practices

- Changing Landscape
  Emerging technologies are altering the market, but:
  - This analysis focuses on conventional fuel vehicles

- Revisiting Assumptions
  - As vehicle design evolves, older assumptions about survival time may no longer hold — especially across different gasoline grades and engine configurations.

Why Vehicle Longevity Matters

- Critical Factor: How long vehicles operate impacts:
  - Environmental footprint (lifetime emissions, resource use)
  - Economic activity (manufacturing, policy, consumer costs)

- Key Characteristics: Fuel type, engine size, emissions shape:
  - Policy decisions (standards, infrastructure)
  - Industry strategy (design, warranty)
  - Consumer choices (cost, value)

Data Overview

Data Collection

- Source: U.S. EPA Fuel Economy Dataset
- Years Covered: 1985–2024
- Sample Size: 47,548 vehicle models
- Censoring: Right-censored for cars newer than 2020
- Hydrogen and Electric vehicles were excluded from our final analysis because there were too few of them in the dataset (especially since we had censoring from 2020 onwards), and many are recent models without enough years on the road to reliably estimate survival time.

Variables

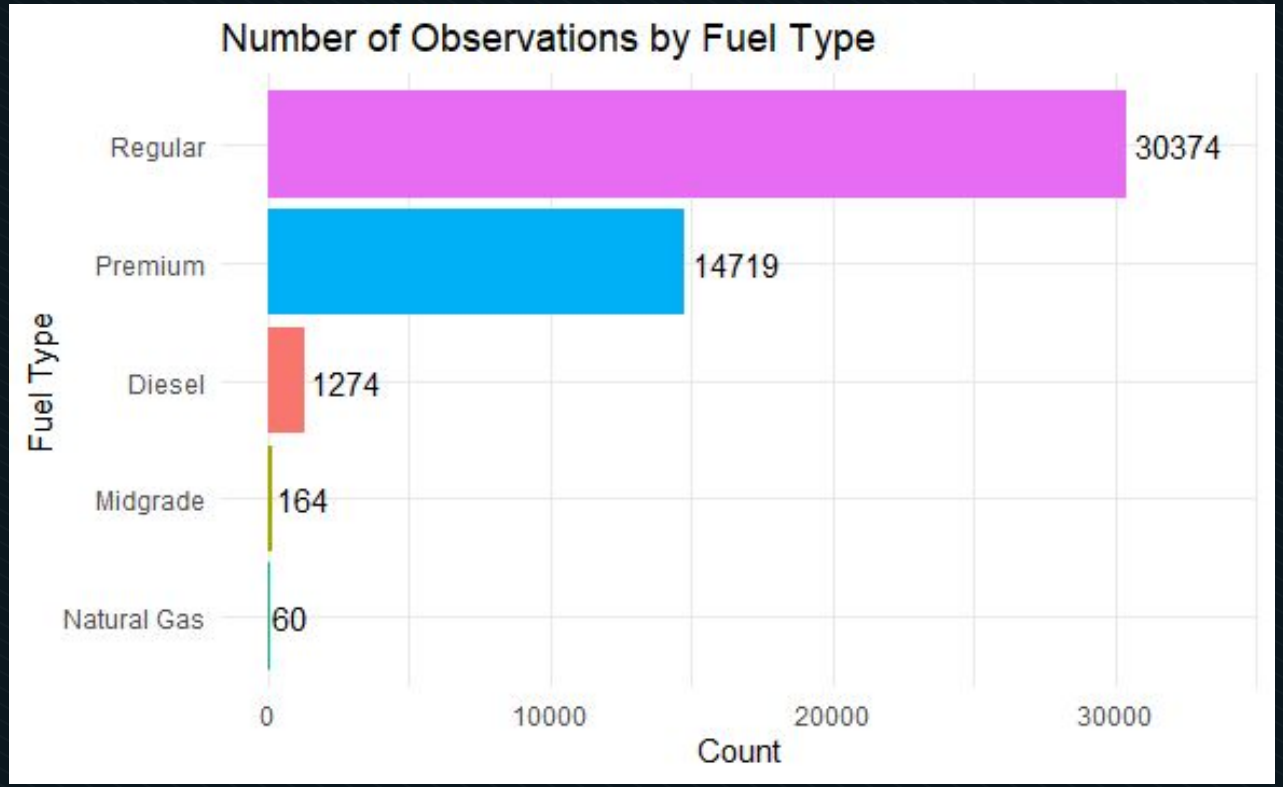
- Response Variable: Time to vehicle removal from use (2025 - model year)

- Covariates: Primary Fuel Type (Regular Gas, Premium, Midgrade, Natural Gas, Diesel

- Experimental Units: Individual vehicle model-year combinations (e.g., 2007 Honda Civic with automatic transmission)

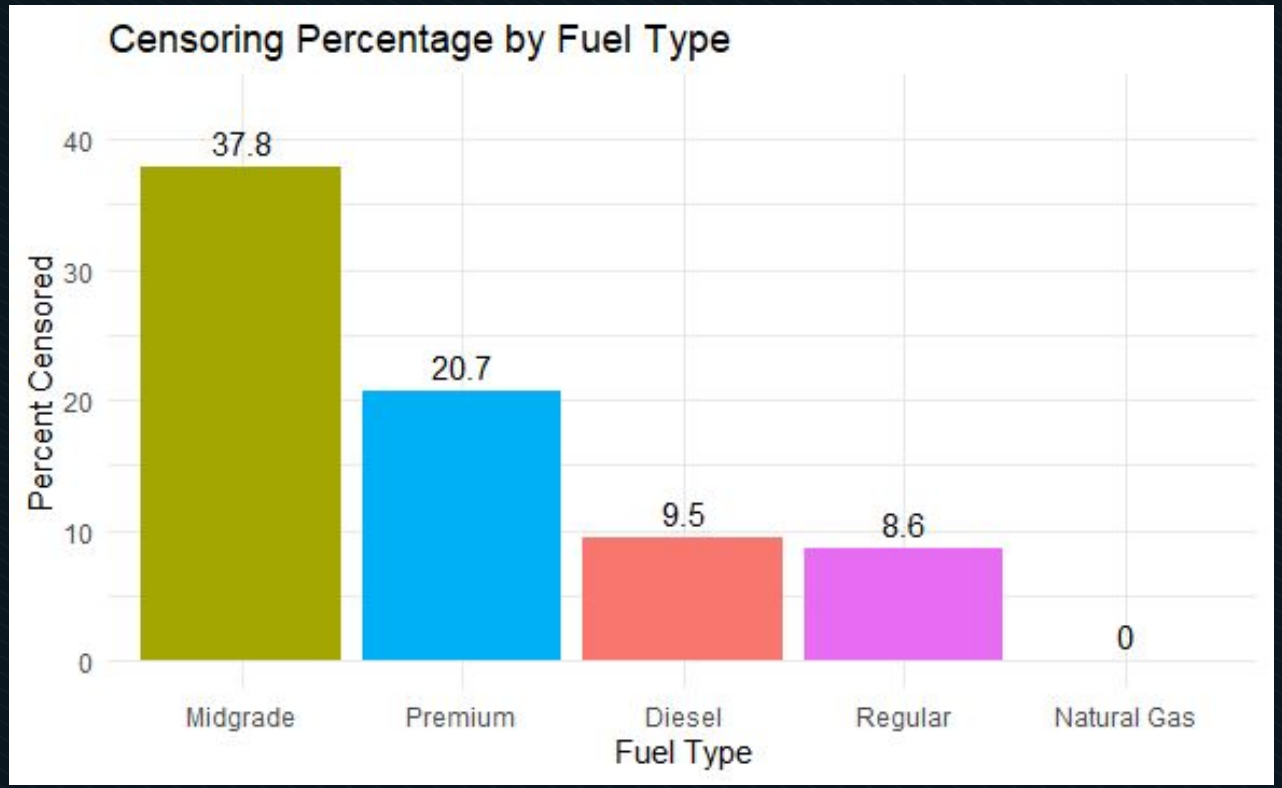
Results and Analysis

Descriptive Statistics



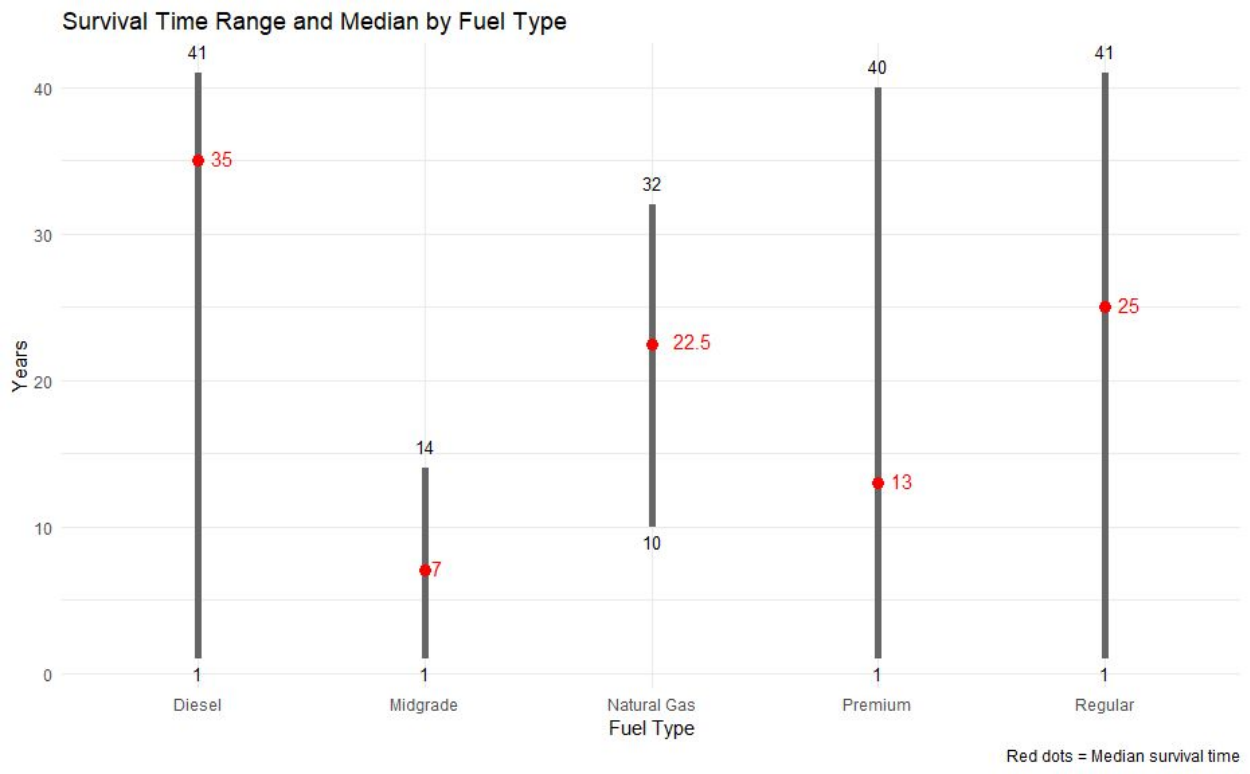
Results and Analysis

Descriptive Statistics



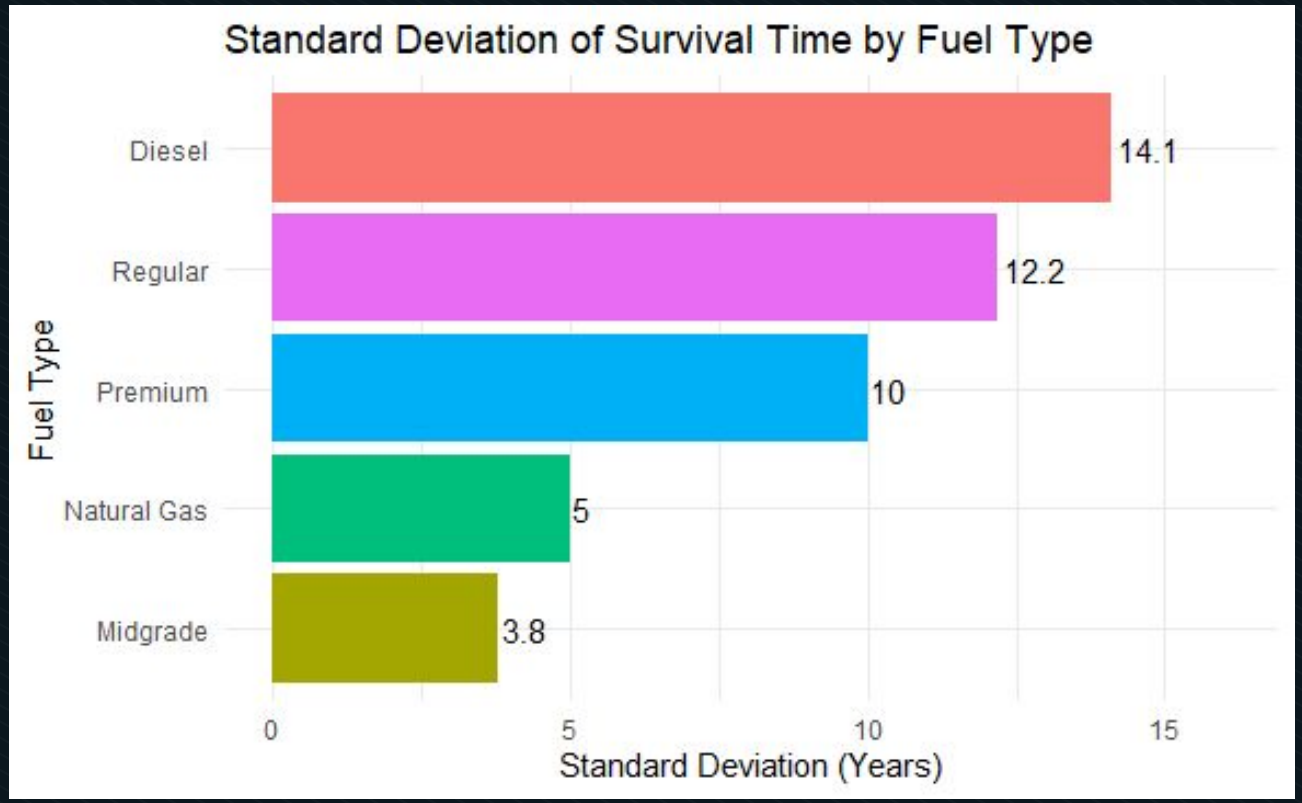
Results and Analysis

Descriptive Statistics



Results and Analysis

Descriptive Statistics



Kaplan-Meier Curve

P-value from log-rank test: 2*10^-16

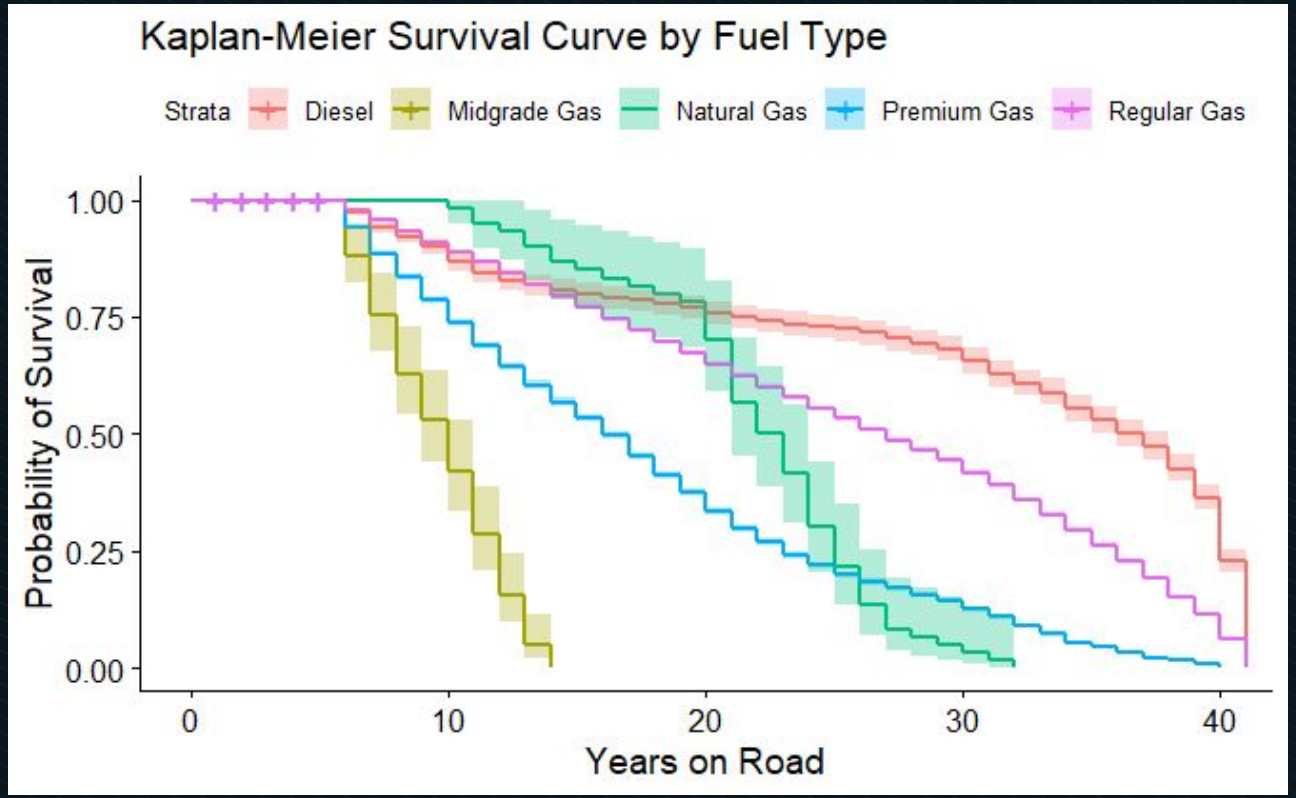
Interpretation: The differences in vehicle longevity by fuel type are statistically significant, meaning that fuel type is a statistically significant predictor of vehicle lifespan.

Diesel vehicles have the highest survival probabilities over time, with many lasting 30+ years

Midgrade Gasoline vehicles have the shortest survival time, with a consistent steep drops after ~7-8 years

Natural Gas and Premium Gasoline vehicles show moderate longevity

Regular Gasoline vehicles fall in the middle but decline steadily



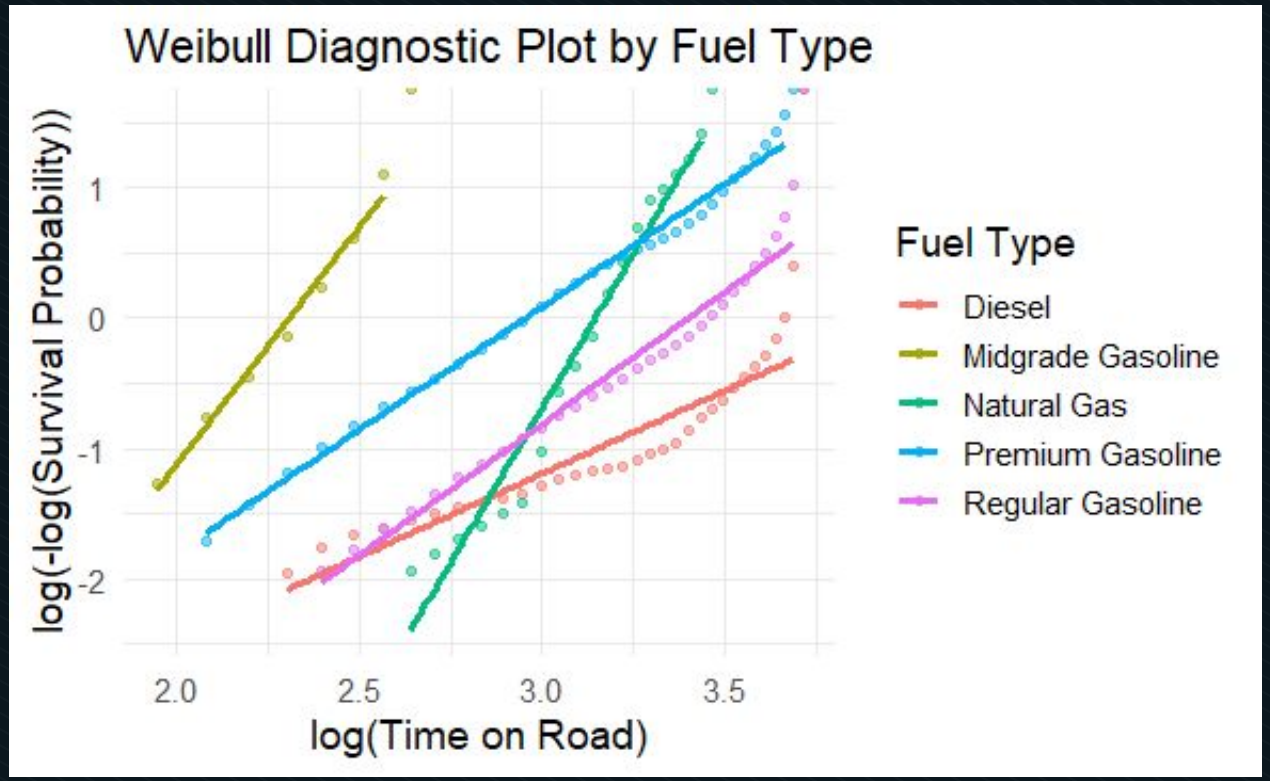
Weibull Diagnostic Plot

Diagnostics & Goodness-of-Fit

- Weibull model chosen based on:
- Lowest AIC (238493.8)
- Weibull plot suggests linearity in log-log survival transformation
- Reasonable parameter estimates

Potential Model Violations:

- Longevity may be influenced by external forces such as poor maintenance (not changing oil frequently, redlining, ect) .
- Amount of censoring varies by fuel type (like how the midgrade gas fuel type had lots of censoring). The data may not completely capture unreported small contributions.

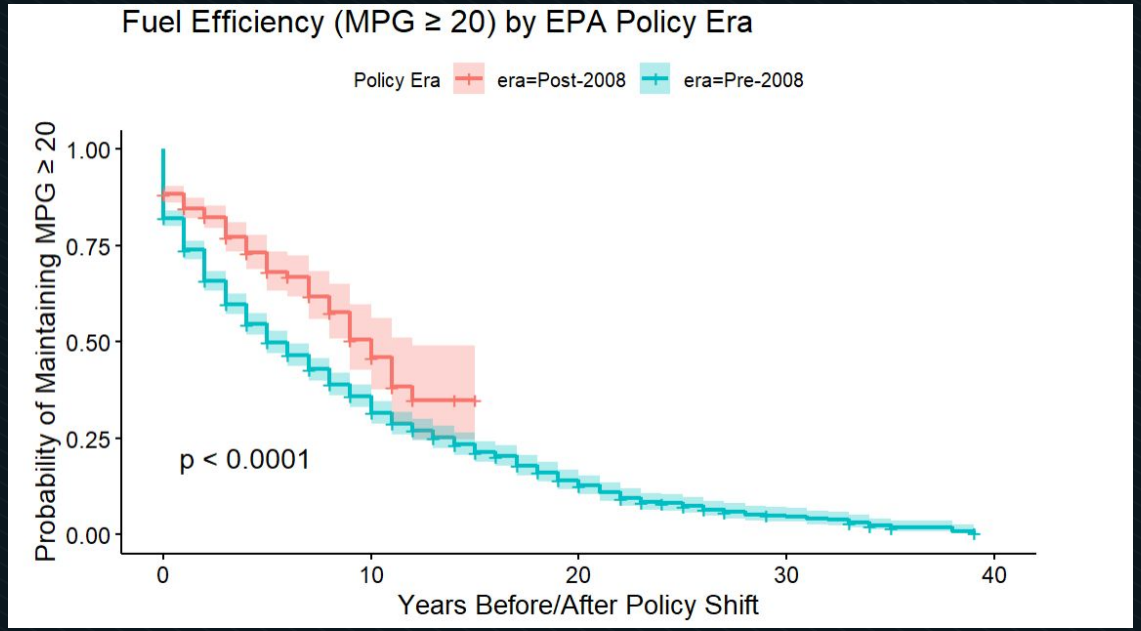


EPA Policy Shift & RFS

2008

- EPA publishes in Federal Register a new renewable fuel standard of 4.66%
- Bush signs Energy Independence ACT (Dec 2007)
- Renewable Fuel Standard Program establishes clear legal criteria for deriving the standard:

$$RFSid_i = 100 \times \frac{RFV_i - Cell_i}{(G_i - R_i) + (GS_i - RS_i) - GE_i}$$



Regression Analysis:

Predicting Fuel Economy

- Turned variables to factors
- Used comb08 as target variable
- Used linear model to predict fuel economy given six factors:
- displ + cylinders + drive + fuelType + VClass + trany

Top 5 Predictors That Increase and Decrease Fuel Economy (MPG)

term	estimate	std.error	statistic	p.value
tranyAutomatic (AM6)	13.681	1.180	11.592	0
tranyAutomatic (AV-S6)	12.270	1.158	10.592	0
tranyAutomatic (AM5)	12.259	1.551	7.906	0
tranyAutomatic (variable gear ratios)	9.923	1.137	8.731	0
tranyAutomatic (AV-S10)	9.915	1.504	6.594	0
VClassSpecial Purpose Vehicle 2WD	-2.861	0.304	-9.403	0
VClassSport Utility Vehicle - 2WD	-2.990	0.128	-23.301	0
VClassVans	-3.153	0.290	-10.873	0
VClassStandard Sport Utility Vehicle 2WD	-3.219	0.467	-6.888	0
VClassMinivan - 2WD	-3.349	0.194	-17.260	0

Most Significant Terms in Predicting Fuel Economy (MPG)

term	estimate	std.error	statistic	p.value
tranyAutomatic (S7)	4.672	1.216	3.842	0.000
tranyAutomatic (S9)	4.676	1.253	3.732	0.000
tranyAutomatic 9-spd	4.190	1.188	3.527	0.000
tranyAutomatic 10-spd	4.257	1.315	3.238	0.001
fuelTypePremium Gas or Electricity	3.756	1.160	3.237	0.001
tranyAutomatic 7-spd	3.550	1.179	3.012	0.003
tranyAutomatic 8-spd	3.427	1.169	2.932	0.003
fuelTypePremium and Electricity	2.550	0.870	2.932	0.003
tranyAutomatic (AM8)	5.511	2.011	2.741	0.006
tranyAutomatic (S6)	3.079	1.143	2.695	0.007
fuelTypeGasoline or E85	1.762	0.660	2.669	0.008
tranyAutomatic 6-spd	2.966	1.146	2.587	0.010
tranyManual 7-spd	4.703	1.867	2.518	0.012
tranyManual 6-spd	2.769	1.145	2.420	0.016
tranyAutomatic (S10)	2.917	1.211	2.409	0.016
tranyAutomatic (AM-S8)	3.526	1.479	2.384	0.017
fuelTypeRegular	1.469	0.645	2.279	0.023
VClassMinivan - 4WD	-1.339	0.589	-2.272	0.023
tranyManual 4-spd	2.334	1.188	1.964	0.050

Conclusions

Main take away

- Fuel type is a strong predictor of vehicle longevity
- Impact on Consumer Choice & Value
- Influence on Fleet Turnover & Environmental Goals

Limitations

- Model year as a proxy for vehicle lifespan
- No actual end-of-life records
- No mileage or usage-based data

Potential future analysis

- Incorporate real-world retirement data (e.g., scrap yard records)
- Add vehicle mileage and maintenance history
- Explore manufacturer effects or technological upgrades
- Consider competing risks (e.g., collision vs. obsolescence)

THANKS FOR WATCHING!!

ANY QUESTIONS?